The Enterprise Growth

A Proposal report for the BDM capstone Project

Submitted by

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<u>Declaration Statement</u>

I am working on a Project Title "The Enterprise Growth". I extend my appreciation to **Omkar Enterprises** for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate: (**Digital Signature**)

Name: Jay Mestry   Date: 3/2/2025

1 Executive Summary of Business

Omkar Enterprises, established in 1996, is a B2B manufacturing company specializing in sheet metal cutting, embossing dies, and power pressing labor work, located in a small village of Maharashtra. The business operates as a proprietorship under Rajesh Mestry, serving diverse industrial clients with custom-made dies and manufactured products.

The company faces significant challenges in three critical areas: production efficiency and delivery timeline management, profit optimization, and order-to-capacity alignment. These challenges stem from traditional manual manufacturing methods, limited machinery capacity, and the need for better production scheduling. The business currently generates an annual revenue of 10-12 lakhs through the manufacturing of various products including pipe clamps, round cutter blades, pan handle attachers, and other industrial components.

The project aims to address these challenges through comprehensive data analysis of production cycles, cost structures, and more. By analyzing historical production data, order fulfillment, and profit margins, this study will develop actionable insights to optimize manufacturing processes, enhance delivery performance, and improve profit margins. The expected outcome will provide strategic recommendations for better resource allocation, production scheduling, and pricing strategies, ultimately strengthening the company's operational efficiency and profitability.

2 Organization Background

Omkar Enterprises, established in 1996 by Rajesh Mestry, is a proprietorship-based manufacturing company located in a small village of Maharashtra. The company specializes in sheet metal cutting, embossing dies manufacturing, and power pressing labor job works, serving a diverse B2B clientele.

Starting with basic machinery including a drill machine, base grinder, surface grinding machine, and a single hand press, the company has grown steadily over the past 27 years. Today, the facility operates with an expanded array of equipment including 5 power press machines, 5 hand press machines, drill machines, base grinder, surface grinding machine, metal sheet cutter, shaper machine, lathe machine, and various manufacturing tools.

The company produces a wide range of industrial components such as pipe clamps, round cutter blades, pan handle attachers, MS washers, machinery livers, keys, wrenches, door hinges, and other specialized products. With an annual revenue of 10-12 lakhs and a workforce of three skilled workers supporting the owner, Omkar Enterprises continues to serve its industrial clients through traditional manufacturing methods and custom die-making expertise.

3 Problem Statement

3.1 Production Efficiency and Delivery Timeline Management: The company faces challenges in meeting delivery timelines and managing production cycles efficiently, leading to delayed order completion and customer dissatisfaction. This impacts the overall operational effectiveness and client relationships.

3.2 Profit Optimization and Cost Management: The business struggles with maintaining optimal profit margins due to challenges in pricing strategies and cost control, affecting its financial sustainability and growth potential.

3.3 Order-to-Capacity Optimization: The company experiences difficulties in balancing incoming orders with available production capacity, resulting in resource allocation inefficiencies and production bottlenecks that affect business performance.

4 Background of the Problem

The identified problems at Omkar Enterprises stem from both internal operational constraints and external market challenges that have evolved since its establishment in 1996.

Internal Factors: The production efficiency and delivery timeline issues primarily arise from the company's reliance on traditional manual manufacturing methods. The manual creation of die drawings and production processes, while skillful, leads to longer production cycles and potential delays. The limited machinery capacity, with only 5 power press machines and 5 hand press machines, creates bottlenecks in production scheduling.

External Factors: The profit optimization challenges are influenced by market competition and pricing pressures in the B2B manufacturing sector. The company's small-scale operations and limited negotiating power affect its ability to command optimal prices for products and dies.

Operational Impact: The order-to-capacity misalignment is a compound effect of these factors. Without automated production planning and capacity management systems, the company struggles to efficiently allocate resources across multiple orders. This results in:

- Suboptimal production scheduling
- Inconsistent delivery timelines
- Resource utilization inefficiencies
- Challenges in managing worker productivity

These problems affect the company's overall operational efficiency and profitability.

5 Problem Solving Approach

My problem-solving approach will focus on three main areas using data collection and analysis methods. For the production efficiency problem, I will start by data in Excel sheets to track when raw materials arrive and when finished products are delivered to clients. By calculating the time taken for each order's completion, I can easily identify which products take longer to manufacture and where the common delays occur in the production process. This information will help me create better production schedules.

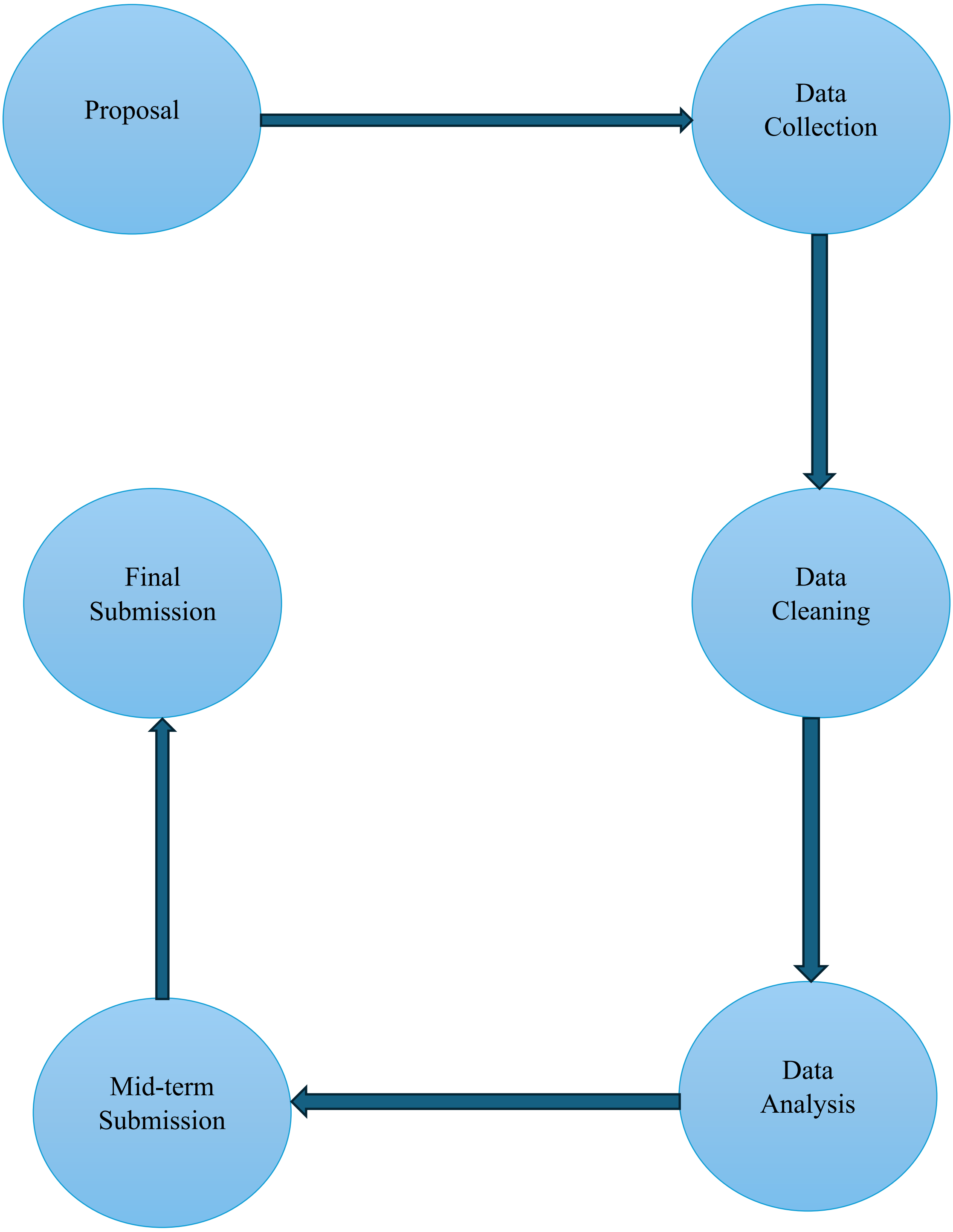
To address profit optimization, I will maintain detailed records of costs and selling prices for each product type. I will separately track both regular product manufacturing and die manufacturing costs. By calculating profit margins for different products, I can make straightforward comparisons to identify which products generate better profits. This analysis will help me determine which products might need price adjustments to improve overall profitability.

Data analysis approach will focus on following key areas to optimize capacity. First, I will analyze production timelines by comparing raw material arrival dates with completion dates and tracking required pieces against completion times. Then, I will examine worker efficiency by studying labor distribution and costs. Then I plan to analyze order patterns by studying purchase dates and comparing new versus existing client orders across different industry types. This comprehensive analysis will help me understand production capabilities, identify time-consuming products, optimize worker assignments, and prioritize orders effectively, leading to better production scheduling and order acceptance decisions.

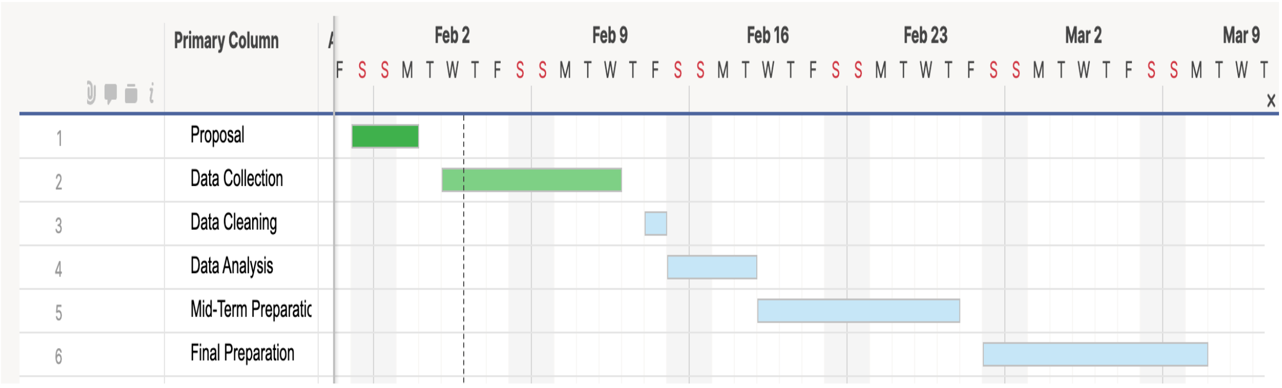
The implementation will follow four simple steps. First, I will create basic Excel sheets collecting 8 months to 1 year of data of year 2024, I will analyze the patterns using simple charts and visualizations. Finally, I will suggest practical improvements based on my findings. The entire approach focuses on using basic data analysis tools to find straightforward, implementable solutions that the business can easily adopt without requiring complex statistical methods or advanced techniques.

6 Expected Timeline

6.1 Work Breakdown Structure



6.2 Gantt Chart



7 Expected Outcome

1. Production Timeline Improvement: By analyzing the time between raw material arrival and product delivery, I will identify which products face consistent delays. This will help create realistic delivery schedules that the business can actually follow, leading to better client satisfaction.
2. Better Profit Understanding: Through analysis of production costs and selling prices across different products and dies, I will determine which items make more money for the business. This will help make better decisions about which products to focus on and what prices to charge.
3. Smarter Order Management: By studying order patterns and worker assignments, I will create a simple system to decide which orders to accept and when. This will help the business avoid taking too many orders at once and ensure they can deliver products on time.